WORKING LOCAL PRODUCT

# ClipTrail

Private, scene-aware search for local video

## Describe a scene. Jump to the mom
## Keep every video local.

**Natural-language query**

“A red-haired woman surrounded by burning rubble.”



**Sintel · 23.25 s · saved local result**

Frame: Blender Foundation / Sintel (CC BY 3.0)

PROBLEM

# The moment is buried inside the file

A filename cannot describe every visual event in a video.

### Manual scrubbing

Finding one moment means opening files and searching the timeline by eye.

### Cloud tradeoff

Remote indexing requires media upload and ongoing external storage.

**The user wants a timestamped moment — with the source kept under local control.**

SOLUTION

# Index scenes once. Search them repeatedly.

A compact local visual index turns a plain-language description into a playable moment.

**01**

## Select

One video or a folder

**02**

## Index

Representative scene frames

**03**

## Search

Natural-language CLIP query

**04**

## Play

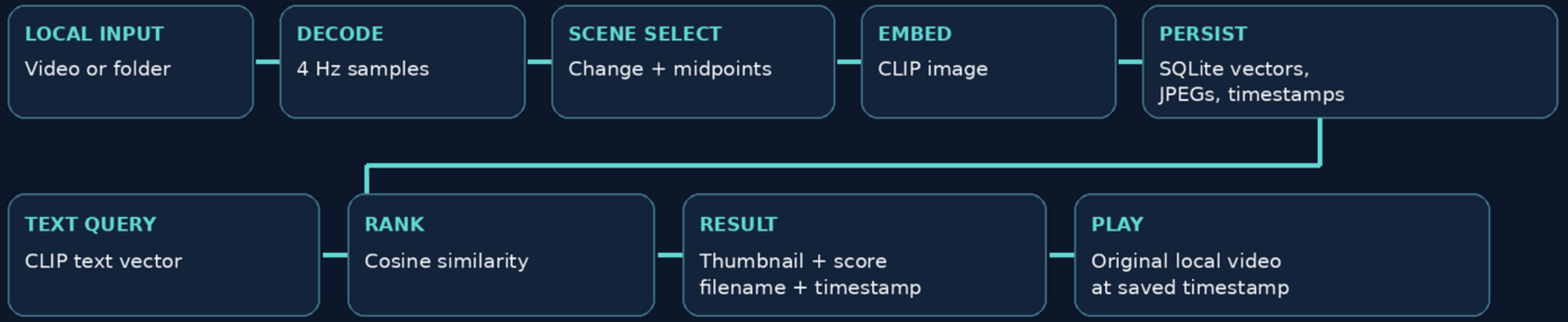
Ranked timestamped result

**Videos, embeddings, queries, and playback stay on the same computer.**

ARCHITECTURE

# Sparse, persistent, local

The verified CPU application path and experimental QNN image path are distinct.




**EXPERIMENTAL IMAGE BACKEND · SEPARATE EVIDENCE SCOPE**

Windows ARM64 → ONNX Runtime QNN → pinned HTP backend → provider-level graph execution

Observed: provider loading, HTP loading, QNN partition, embedding parity.

**Physical NPU device execution, acceleration, and full target application deployment remain unverified.**

EVIDENCE

# Measured on the recorded CPU host

Frozen retrieval benchmark and persistent application diagnostics have different scopes.

**87.50%**

**Recall@3**

24 positive queries · 2 films

**81.82%**

**Short-event R@3**

11 positive queries

**27.45 ms**

**Warm search median**

5 repeat queries · prebuilt index

**6.80 s**

**Index / video minute**

After model load

**~250 KB**

**Logical index / minute**

Vectors + JPEG thumbnails

CPU: AMD Ryzen 5 5600H · WSL2/Linux · one recorded three-video application run

SNAPDRAGON / QNN

# Provider-level compatibility observed

One fixed CLIP image graph smoke on an ASUS Snapdragon X2 Elite.

## Observed

Native ARM64 graph execution; QNN provider and pinned HTP libraries loaded; one QNN partition; CPU-reference embedding parity (cosine 0.999982834).

## Unverified

Physical NPU device execution; speedup or energy advantage; production stability; full ClipTrail or HP-device deployment.

**Hardware profiler device events were absent. The backend remains experimental.**

CLOSING

# Private video becomes searchable

A working local product with measured sparse retrieval and a precise future ARM64/QNN path.

**Describe a scene.**

**Jump to the moment.**

**Keep every video local.**

Working now: persistent CPU application • Future work: broader evaluation and device-level QNN proof